Asian Semis

August WSTS: Logic growth accelerates on AI compute momentum; memory maintains elevated pricing

Overall semi revenues rose 156% YoY in August (vs 131% YoY in July), with logic semis increasing 45% YoY (vs 37% YoY in July) and memory increasing 412% YoY (vs 435% YoY in July). Key drivers include: 1) robust logic momentum supported by surging AI accelerator and CPU demand, with continued capacity expansion, broader price hikes in 2027, and TSMC indicating AI demand visibility now extending into 2029-30; 2) memory's sustained strength underpinned by structurally tight HBM supply-demand through 2028E; and 3) continued AI capex momentum, with major US hyperscalers reporting accelerating cloud revenue growth and leverage ratios remaining low at 13% net debt to equity.

Logic semi revenue rose 45% YoY (vs 37% in July), marking 36 consecutive months of YoY growth. On leading edge, N3 supply-demand tightness is expected to worsen in 2027 as major AI accelerators remain on N3 (NVIDIA Rubin, Google TPU v8, AWS Trainium 3). N2 capacity is ramping faster than prior expectations (reaching 118k/210k wfpm by end-27/28) to support flagship accelerator and CPU programs in the coming years. We now expect TSMC capex to reach $86/$100bn in 2027/28 (from prior $81/$90bn), with 9 leading-edge fab phases expected to come online over the next 3 years for N2. For mature foundries, pricing momentum continues to strengthen in 8" capacity given AI-peripheral applications, with further upside expected in 2027.

Overall semi units increased 19% YoY (vs 16% YoY in July). Memory units increased 65% YoY (vs 80% YoY in July), and logic semi units rose 16% YoY (vs 14% YoY in July). Within memory, ASIC is expected to overtake NVIDIA as the largest HBM consumer in 2027 (ASIC 48% vs NVIDIA 43% of total HBM bit demand), driven by substantial ASIC system unit growth (+102% YoY vs NVIDIA +15%). We continue to expect HBM wafer allocation to reach 31% of total DRAM capacity by 2028E-end, with 58% of the incremental DRAM capacity (930k WSPM increase from 2025-end to 2028-end) allocated to HBM-grade front-end wafer fabrication. Logic unit growth was supported by continued AI compute demand across accelerators, datacenter CPUs, and networking, partially offset by weak consumer electronics.

Overall semi ASPs were up 116% YoY (vs 99% YoY in July), with memory ASPs rising 210% YoY (vs 198% YoY in July). Memory ASPs continue to accelerate on structurally tight supply-demand and increasing HBM mix within shipments. We have revised up our HBM ASP growth assumptions to 54%/25% in 2027E/28E (vs prior 43% for 2027E), reflecting tighter-than-expected supply-demand. For DRAM, we expect ASP to rise 29%/7% in 2027E/28E, while for NAND, we expect ASP to rise 23% in 2027E before declining 4% in 2028E due to the diverging price trend between eSSD and consumer-grade SSD. Logic semi ASPs rose 25% YoY (vs 21% YoY in July), supported by continued mix shift towards advanced nodes (higher N3 mix and early N2 ramp) and sustained 100%+ utilization at N5/N3. We expect broader TSMC price hikes in 2027 (~10% for N3/N2, low-to-mid single digit for mature nodes (N7 and above) – the first mature node increase in this upcycle), alongside strengthening 8" pricing at mature foundries driven by AI-peripheral demand.

Technology and Telecoms

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Looking ahead, we do not see fundamental indicators that signal meaningful weakness in the AI-driven semiconductor upcycle. The AI economics remain robust, with frontier model vendors achieving 60-80% inference gross margins and generating $20-40bn in annual revenue per GW (vs $10bn in 2025). Hyperscaler capex sustainability is supported by low leverage ratios (aggregate net debt/equity at ~13%) and accelerating public cloud revenue growth (+43% YoY in 2Q26), with aggregate CSP capex forecast to grow 65% in 2027. On the logic side, we expect TSMC to deliver 39% revenue growth in 2027, driven by N3 tightness and accelerating N2 ramp, maintaining 95%+ market share in N2/A16. Mature foundries should see continued ASP improvement and sustained AI-peripheral demand. On memory, HBM demand is expected to grow at 63% CAGR through 2028E, with ASIC expected to overtake NVIDIA as the largest consumer in 2027, and HBM TAM reaching $160bn/$282bn in 2027E/28E. We expect S-D shortage to continue with the degree of shortage worsening in 2027E followed by a slight improvement into 2028E.

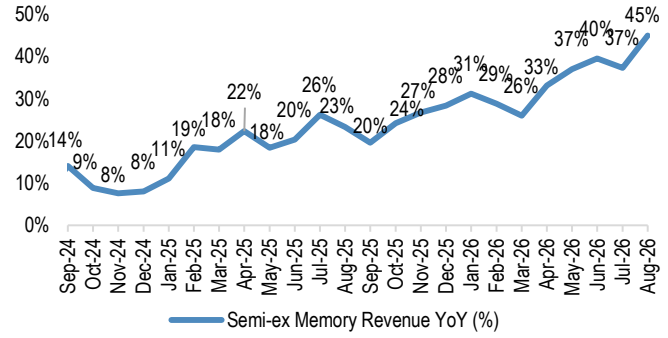
Stock picks: We reiterate our top picks: TSMC, ASE, Unimicron, Aspeed, Hon Precision, SKH, Chroma, Advantest, and Tokyo Electron.

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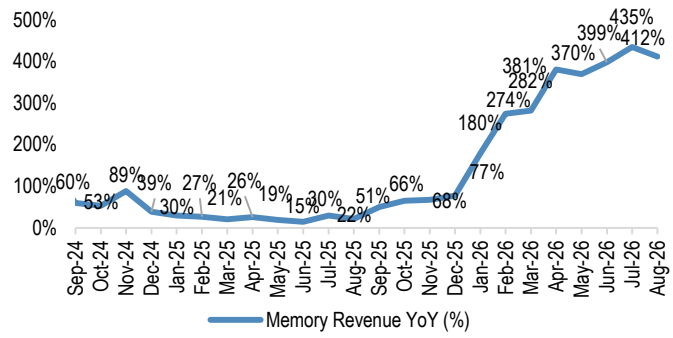
Key charts

Figure 1: Semi ex-memory revenue YoY (Sep-2024 onwards)



Source: WSTS.

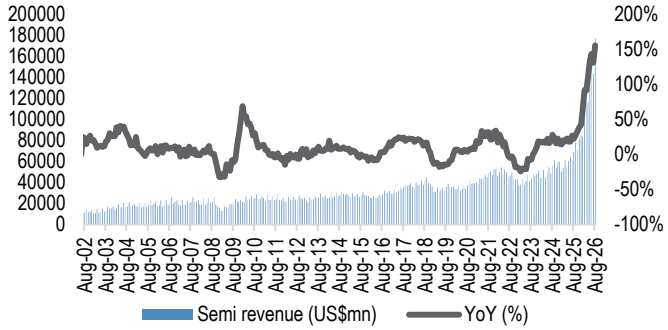
Figure 2: Memory revenue YoY (Sep-2024 onwards)



Source: WSTS.

Figure 3: Semi revenue (monthly) and YoY

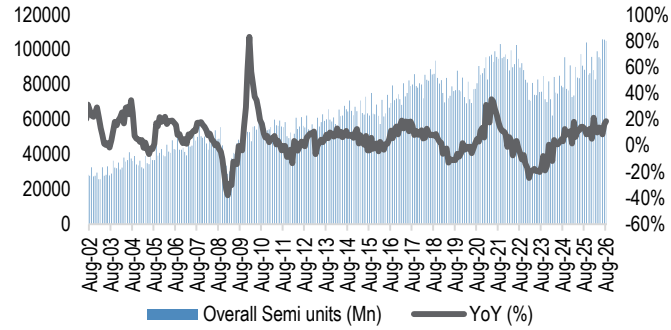
US$mn, %



Source: WSTS.

Figure 4: Semi units (monthly) and YoY

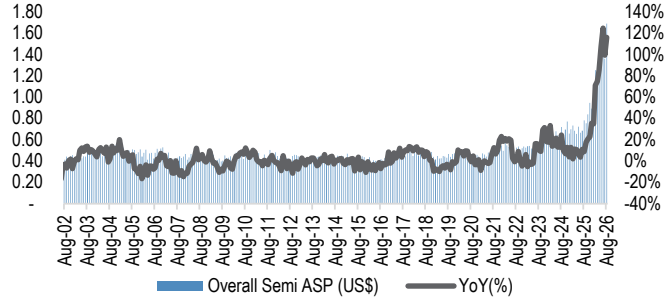
mn, %



Source: WSTS.

Figure 5: Semi ASP and YoY

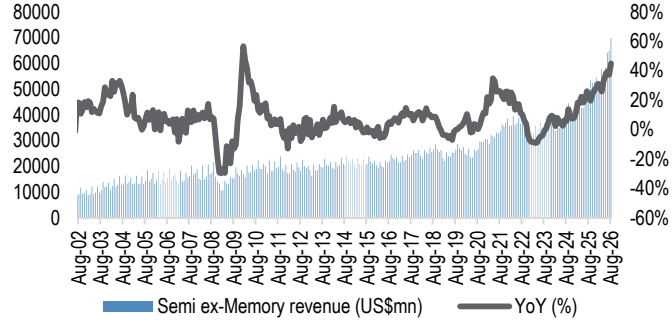
US$, %



Source: WSTS.

Figure 6: Semi ex-memory revenue (monthly) and YoY

US$mn, %

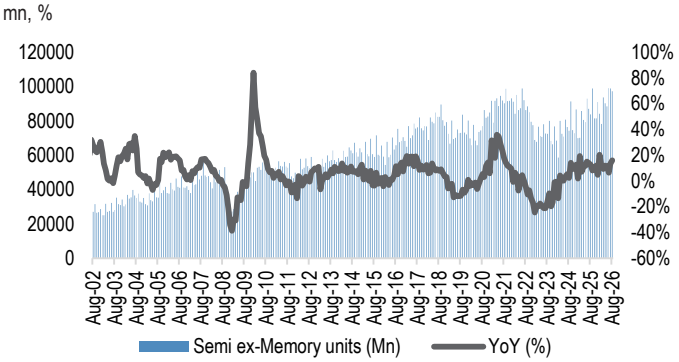


Source: WSTS.

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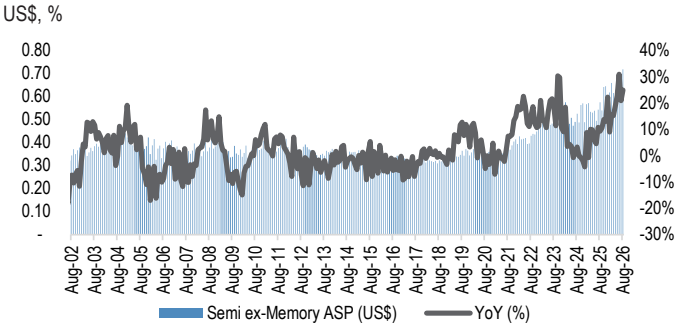
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Figure 7: Semi ex-memory units and YoY



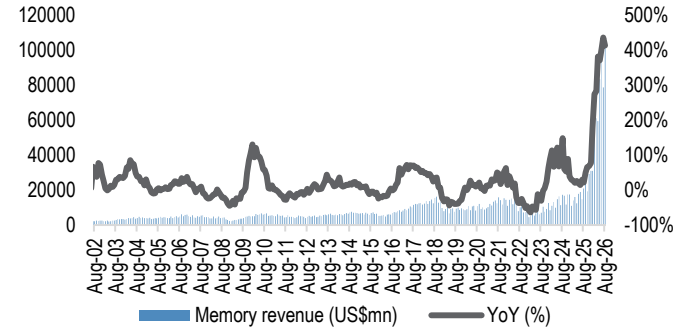
Source: WSTS.

Figure 8: Semi ex-memory ASP and YoY



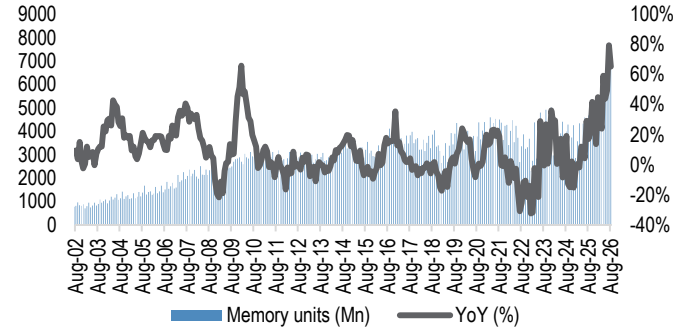
Source: WSTS.

Figure 9: Memory revenue (monthly) and YoY



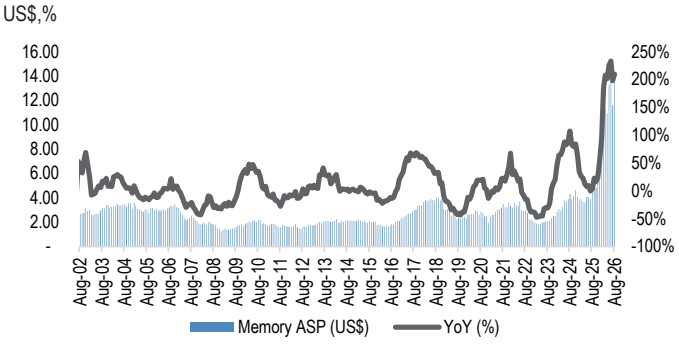
Source: WSTS.

Figure 10: Memory units and YoY



Source: WSTS.

Figure 11: Memory ASP and YoY



Source: WSTS.

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SK hynix (000660.KS, 000660 KS) Price Chart



Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
Initiated coverage Dec 31, 2000. All share prices are as of market close on the previous business day.

Date	Rating	Price (W)	Price Target (W)
07-Nov-23	OW	133000	160,000
25-Jan-24	OW	141500	175,000
22-Feb-24	OW	149000	185,000
28-Mar-24	OW	181200	260,000
16-Jul-24	OW	230000	300,000
08-Sep-24	OW	156400	240,000
24-Oct-24	OW	196000	260,000
10-Dec-24	OW	168900	210,000
23-Jan-25	OW	225500	280,000
03-Apr-25	OW	197900	300,000
24-Apr-25	OW	181000	285,000
25-Jun-25	OW	278500	360,000
24-Sep-25	OW	361000	460,000
23-Oct-25	OW	481000	650,000
29-Oct-25	OW	519000	700,000
14-Dec-25	OW	571000	800,000
16-Jan-26	OW	756000	1,000,000
29-Jan-26	OW	862000	1,250,000
22-Mar-26	OW	1006000	1,550,000
23-Apr-26	OW	1220000	1,800,000
17-May-26	OW	1848000	3,000,000
05-Aug-26	OW	1609000	2,750,000

Tokyo Electron (8035) (8035.T, 8035 JP) Price Chart



Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
Initiated coverage Apr 04, 2001. All share prices are as of market close on the previous business day.

Date	Rating	Price (Y)	Price Target (Y)
09-Feb-24	OW	5887	6,840
09-Jul-24	OW	7208	8,200
11-Dec-24	OW	4920	7,000
13-Jun-25	OW	5000	6,200
12-Sep-25	OW	4281	5,200
08-Nov-25	OW	6560	7,600
10-Jan-26	OW	7582	9,000
13-Mar-26	OW	7950	10,200
17-Jun-26	OW	14172	17,000

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